Journal of Hydrology 658 (2025) 133192

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Time-varying mechanisms of hydraulic properties of root-soil composites under plant root decay

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A R T I C L E I N F O

This manuscript was handled by Renato Morbidelli, Editor-in-Chief, with the assistance of Vincenzo Bagarello, Associate Editor.

Keywords:
Root decay
Time-varying model
Hydraulic properties
Evolution mechanism
Soil structure

A B S T R A C T

The hydraulic effect of plant roots reduces precipitation infiltration and enhances shallow slope stability. However, after root death and decay, soil permeability increases while water-retention capacity decreases. The time-varying mechanisms governing the hydraulic properties of root-soil composites after root decay remain unclear. This study examines the evolution of soil pore structure following root decay. A time-varying soil water retention curve (SWRC) model was developed to characterize changes in water-retention capacity. Additionally, a time-varying saturated infiltration coefficient model and a permeability coefficient prediction model were established to describe variations in hydraulic properties. A one-dimensional soil column infiltration test was conducted on root-soil composites at different stages of root decay to investigate the time-dependent changes in hydraulic properties. The reliability of the proposed models was validated using experimental results.

The findings indicate the following: After root death, root biomass, diameter, length, and number decreased with increasing decay time, stabilizing after four months. Root decay led to a reduction in root volume ratio, which altered soil structure and enhanced the permeability of root-soil composites. Longer decay periods increased soil porosity, modifying the soil water characteristic curve and reducing water-retention capacity. Creeping roots decayed more significantly than fibrous roots due to their distinct morphological traits, making changes in hydraulic properties more pronounced in the topsoil. Therefore, plant root decay negatively affects soil hydraulic properties by continuously altering soil pore structure. These findings provide a crucial foundation for understanding the time-dependent mechanisms of hydraulic property variations in root-soil composites during plant root decay.

1. Introduction

The hydraulic properties of soil are improved by the plant roots (Ng et al., 2015). During plant transpiration, root water uptake transfers water from the soil to the atmosphere, reducing soil water content and permeability while increasing suction (Nyambayo and Potts, 2010; Ng et al., 2016; Bertolini et al., 2023). This process improves the soil's water-retention and impermeability capacity (Zhou et al., 2025).

In recent years, land degradation has resulted in widespread plant death, reduced surface plant cover, and severe soil erosion (Wang et al., 2022; Srivastava and Bharti, 2023). The frequency of heavy precipitation has increased due to climate change, and land without plant cover is more prone to secondary geological disasters such as shallow landslides and debris flows (Zhou and Qin, 2022; Guo et al., 2023; Jurchescu et al., 2023). Previous studies have shown that plants enhance soil water-retention capacity and decrease the permeability capacity, which is related to the occupation of soil pores by the roots (Ni et al., 2020). Healthy plant roots bind the soil particles together in a network so that the soil has a good aggregate structure and optimal pore conditions (Xiao et al., 2020; Liu et al., 2022).

However, after plant death, root decay due to microbial activity leads to a reduction in root biomass, diameter, and length, altering soil structure and hydraulic properties (Kamchoom et al., 2022; Lei et al., 2022). Specifically, as root diameters decrease, numerous macropores form between the root and the soil, loosening the soil structure (Xin et al., 2016; Jiang et al., 2018). Once root xylem decays, cavity channels develop, resulting in large interconnected macropores in the soil (Ghestem et al., 2011). During precipitation events, water is more likely to infiltrate into these macropores (Wei et al., 2021), and the consequent seepage dislodges soil particles, forming preferential flow pathways

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https://doi.org/10.1016/j.jhydrol.2025.133192
Received 18 January 2025; Received in revised form 14 March 2025; Accepted 24 March 2025
Available online 27 March 2025

(Balfour et al., 2014; Zhang et al., 2015). This process accelerates the soil infiltration rate (Ni et al., 2024).

Dye tracer tests have shown that decayed roots can significantly increase deep soil water content, increase the water content of deep soil (Wu et al., 2017; Guo et al., 2019), and affect the hydraulic properties of deeper soil (Xiao et al., 2024).

Changes in the hydraulic properties of root-soil composites due to root decay have received little attention in existing infiltration analyses. Only Ni et al. (2019) proposed a soil water retention curve (SWRC) model and a saturated permeability coefficient (k_s) model for vegetated soils to account for root decay. Notably, the degree of root decay gradually increases over time (Vergani et al., 2017; Lei et al., 2022; Wang et al., 2022), leading to the evolution of soil pores and corresponding changes in hydraulic properties. However, existing models fail to capture the dynamics of root decay and cannot predict changes in permeability and water-retention capacity at different stages of decay.

The objectives of this study were to: (1) Derive a time-varying model for hydraulic parameters based on soil pore evolution laws during decay. (2) Determine the relationship between plant root morphological traits (root number, length, diameter, biomass, and volume) and decay time. Develop a root volume ratio decay equation and identify related parameters. (3) Conduct one-dimensional soil column infiltration tests to examine variations in soil permeability and water-retention capacity over time and investigate the underlying causes of these changes to validate the time-varying evolution. This research will enhance our understanding of the mechanisms behind secondary geological disasters, such as landslides triggered by plant death. (4) The reliability of the time-varying models for hydraulic parameters using experimental results, providing a theoretical basis for understanding the evolution of hydraulic properties in root-soil composites due to root decay.

2. Model formulation

2.1. SWRC time-varying model

The roots occupy the pore volume of the soil during growth, and Ng et al. (2016) established the relationship between the void ratio e and the root volume ratio R_v of the vegetated soil based on the soil three-phase theory. However, after plant death, roots gradually decay and decompose into humus, and the channels formed by root decay cause the pores in the root-soil composite gradually increase (Ni et al., 2019). By introducing the root volume ratio (volume of plant roots within a unit volume of soil) decay equation $R_v(t)$ in the $e \sim R_v$ model developed by Ng et al. (2016), the root-soil composite void ratio at the moment of root decay t can be expressed as:

$$e_t = \frac{e_0 - (1+e_0)R_v(t)}{1+(1+e_0)R_v(t)} \tag{1}$$

Where e_0 is the void ratio of bare soil, and $R_v(t)$ is the root volume ratio at different decay times (cm^3/cm^3), which can be determined based on experimental data on root volume ratio at different stages of decay time t, and analyse the evolutionary law of the root volume ratio allows for the derivation of the root volume ratio decay equation $R_v(t)$ which will be discussed later.

Ni et al. (2019) modified the pore-ratio-dependent SWRC model proposed by Gallipoli et al. (2003) to account for the reduction in air-entry value (AEV) due to the formation of large pores in the root-soil composite as a result of root decay:

$$S_r = \left[1+\left(\frac{se^{m_4}}{m_3\exp(-p\frac{e_r}{e_{r0}})}\right)^{m_2}\right]^{-m_1} \tag{2}$$

where Sr is the degree of saturation, s is the matric suction (kPa), and e is the soil void ratio. The parameters α,m_1,m_2,m_3, and m_4 are model-fitting parameters, where m_1 and m_2 control the shape of the SWRC, while m_3 and m_4 are related to the AEV. The parameters m_2,m_3, and m_4 are multiplied by 1. The p is the model parameter controlling the rate of reduction in air-entry value due to root decay and is influenced by plant and soil type. The e_r is the void ratio of the volume of soil particles occupied by pores formed through root decay, while e_{r0} denotes the void ratio of the volume of soil particles occupied by roots before decay.

Considering the variation in roots over time, the relationship between e_r and root decay time is expressed as $e_r(t)$:

$$e_r(t) = \frac{(R_{v0} - R_v(t))(1+e_0)}{1+(1+e_0)R_v(t)} \tag{3}$$

where the root volume ratio of the undecayed root is R_{v0} (cm^3/cm^3); at this point decay time $t = 0$, $R_v(t) = R_{v0}$ (i.e. $e_r(t) = e_{r0}$).

Therefore, the SWRC equation (Eq. (2) under root decay is considered, and the root-soil composite pore evolution model (Eq. (1) along with e_r(t)(Eq. (3) is introduced to establish a time-varying SWRC model:

$$S_r = \left[1+\left(\frac{se_t^{m_4}}{m_3\exp(-p\frac{e_r(t)}{e_{r0}})}\right)^{m_2}\right]^{-m_1} \tag{4}$$

By comparing Eq. (4) with Eq. (2) and considering the effect of root decay time on the SWRC, $R_v(t)$ is also introduced. When $t = 0$, Eq. (4) is equivalent to Eq. (2). Thus, by knowing the SWRC of root-undecayed soil and the root volume ratio decay equation $R_v(t)$, the SWRC of the root-soil composite can be predicted at different decay times. Additionally, if the parameter related to $R_v(t)$ is zero, then Eq. (4) represents the SWRC of bare soil.

2.2. Saturated permeability coefficient time-varying model

To establish a time-varying saturated infiltration coefficient model under root decay conditions, this study incorporates the root-soil composite void ratio evolution model e_t (Eq. (1)) into the Kozeny-Carman (KC) model (Kozeny, 1927) to establish a relationship between the saturated permeability coefficient of the soil and the time of root decay, $k_s(t)$.

The KC model estimates the saturated permeability coefficient, k_s, by analyzing the relationship between the void ratio and soil permeability. Its expression is:

$$k_s = A\frac{e^3}{1+e} \tag{5}$$

where e is the soil void ratio, and A is a composite coefficient that depends on the pore shape, the curvature of the water flow channels, the characteristics of the soil particles per unit volume (e.g., shape, size, roughness), and the unit weight of water.

To account for the role of roots, Ni et al. (2019) modified the parameter A in Eq. (5) to:

$$A' = \frac{A}{\exp(-p\frac{e_r}{e_{r0}})} \tag{6}$$

In this study, the variation in root decay over time was further considered, leading to the development of a time-varying saturated permeability coefficient model under root decay, $k_s(t)$was obtained as:

$$k_s(t) = \frac{Ae_t^3}{(1+e_t)\exp(-p\frac{e_r(t)}{e_{r0}})} \tag{7}$$

where A can be calibrated based on the known e and k_s value of root-undecayed soil, while the remaining parameters retain the same meaning as in Eq. (4). When $t = 0$ (i.e.$A' = $ A), the k_s of the root-soil composite at different decay times can be predicted by using the k_s of root-undecayed soil and the root volume ratio decay equation, $R_v(t)$. Similarly, if the parameter associated with root decay, $R_v(t)$ is 0, at this point, Eq. (7) represents the k_s of bare soil.

2.3. Permeability coefficient prediction model

The permeability coefficient function $k_u(t)$, at different decay times, is expressed using the equation proposed by Van Genuchten (1980) as follows:

$$k_u(t) = k_s(t)S_r^{\frac{1}{2}}\left[1 - \left(1 - S_r^{\frac{1}{m_1}}\right)^{m_1}\right]^2 \tag{8}$$

Where, $k_s(t)$ represents the time-varying saturated permeability coefficient model Eq. (7), and S_r is the time-varying SWRC model Eq. (4).

In summary, when using the time-varying model to predict hydraulic parameters, two key aspects must be considered. First, the root volume ratio decay equation, $R_v(t)$, must be obtained. Second, it is necessary to determine three hydraulic parameters: the SWRC, permeability coefficient curves, and the saturated permeability coefficient of root-undecayed soil. If the parameters related to $R_v(t)$are zero, the model reduces to the hydraulic parameter models of bare soil.

3. Test materials and methods

3.1. Soil type and plant specimen preparation

In this test, root-soil composite specimens were prepared through indoor planting. According to the Unified Soil Classifcation System (ASTM, 2017), the test soil consisted of clayey sand (SC). The soil particle size distribution curve is shown in Fig. 1, and other soil properties are listed in Table 1.

Cynodon dactylon was used in the test due to its well-developed creeping and fibrous roots, which provide a robust root structure and strong soil fixation ability. It is considered an optimal plant for ecological slope protection engineering (Zhang et al., 2013). Root-soil composite specimens were prepared by artificially inducing root decay at different time intervals. The specific procedure was as follows:

(1) The test soil was adjusted to a dry density of 1.3 g/cm^3 and filled into a planting box. To ensure that the plant roots could grow vertically, the soil was filled to a height of 25 cm. After filling, *Cynodon dactylon* was planted, and the specimens were placed in an environment conducive to growth. The plants were watered regularly and in measured amounts every day for three months, with continuous monitoring to ensure uniform growth.
(2) After three months of growth, the 25 cm-high root-soil composite was extracted using a PVC tube (inner diameter: 20 cm, height: 29 cm). The root-soil composites were then artificially killed using an herbicide solution containing 41 % Glyphosate (Kamchoom et al., 2022). The upper grass was cut off, leaving the roots to decay over intervals of 0, 1, 2, 4, and 6 months. The herbicide solution was applied twice a week during the decay period to prevent excessive soil drying out.

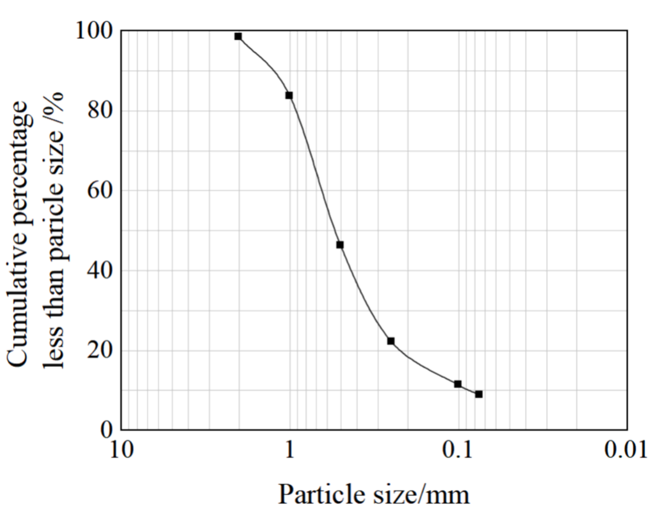

Fig. 1. Soil particle grading curve.

3.2. Design of the soil column infiltration device

A soil column model was used to conduct a one-dimensional infiltration test. The test device consisted of a Plexiglas soil column, a Mariotte bottle, and a data acquisition device (using G0 as an example) (Fig. 2a–b).

The inner diameter of the soil column model was 20 cm, and the total soil height in the column was 40 cm. This included a 25 cm layer of root-soil composite with varying decay times in the upper portion and a 15 cm layer of bare soil in the lower portion. To prevent the formation of preferential flow, a uniform coat of Vaseline was applied to the inside of the model before filling the soil columns.

During the filling process, the lower bare soil layer was compacted to a dry density of 1.3 g/cm^3. The soil was added in 3 cm layers, with each layer consisting of 1.22 kg of pre-weighed soil. After placing each layer, the soil was tamped evenly, and the surface was scraped smooth. This process was repeated five times to achieve a total 15 cm height of the bare soil layer.

The upper root-soil composite was connected to the soil column using PVC pipes to prevent radial deformation during filling. Throughout the test, periodic observations were made to check for the presence of preferential generated.

The calibrated electronic tensiometers and soil moisture sensors were inserted symmetrically into the soil at depths of 5 cm, 20 cm, and 30 cm. They were then sealed with water-stopping tape and sealing mud to prevent water leakage. The electronic tensiometer had a measurement range of 0–90 kPa, with an accuracy of $\pm$ 0.2 kPa. The soil moisture sensor had a measurement range of 0–100 %, with an accuracy of $\pm$ 2 % within the 0–50 % range and $\pm$ 3 % within the 50–100 % range.

During the test, the surface of the soil column was uniformly watered with a precipitation intensity of 20 mm/h for 3 h. Percolating water drained through the porous base plate, flowed into the spherical funnel beneath the base, and collected in a beaker. The amount of percolating water was measured every 5 min using an electronic balance with an accuracy of 0.01 g. The test was terminated when the difference in the amount of percolated water was less than 5 g.

In this experiment, the root-soil composite grown for 3 months without decay was used as the control group (G0). Root-soil composites with root decay times of 1, 2, 4, and 6 months were designated as D1, D2, D4, and D6, respectively, for a total of five test groups.

3.3. Measurement and calculation of root morphological traits

Cynodon dactylon has two types of roots: creeping and fibrous. The fibrous roots grew vertically, while the creeping roots grew horizontally (Fig. 3a). Since the two root types exhibited different growth forms and occurred at different depths (Fig. 3b), the root morphological trait of each type were recorded after decay.

At the end of the infiltration test, the decayed roots were carefully removed from the surrounding soil and immersed in water. The water was gently agitated to separate the soil from the roots. The decayed roots were then retained by filtering through a sieve with a 0.5 mm pore size. After absorbing excess moisture from the root surfaces using a paper towel, the creeping and fibrous roots were separated, and the number of roots was exhaustively counted. The root diameter was measured using a caliper (measurement accuracy: 0.01 mm), while the root length was measured using a straight ruler. The roots were then dried in an oven at 60 °C until they reached a constant weight to determine root biomass (Frasier et al., 2016).

The AEV was calculated to indicate the rate of root decay. To avoid deviations in the root decay constant k caused by variations in initial

Table 1
Basic properties of soil used in the test.

Soil class	Optimum water content (%)	Maximum dry density (g·cm^{-3})	Liquid limit (%)	Plastic limit (%)	Ip	Cu	Cc
clayey sand	22.1	1.65	42.3	30.7	11.6	6.47	2.47

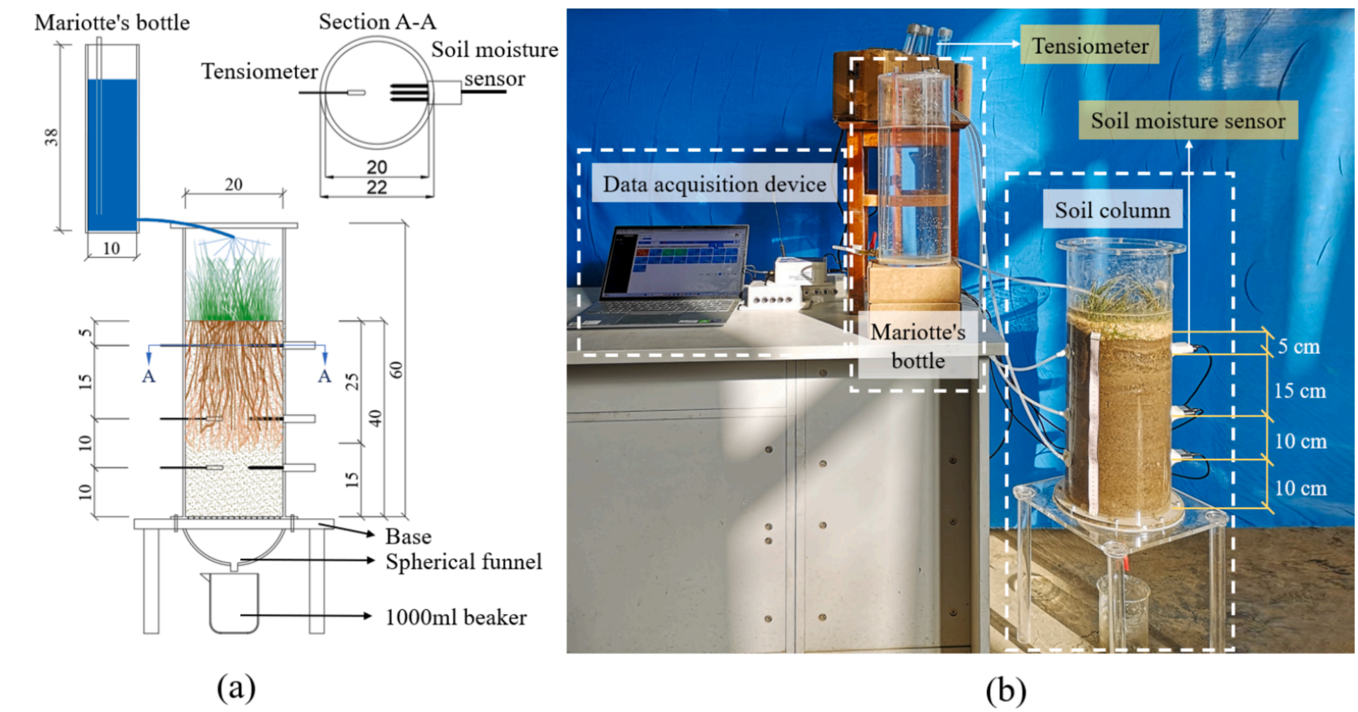

Fig. 2. One-dimensional soil column infiltration test (a) Schematic diagram of soil column test (unit: cm) (b) Soil column test device.

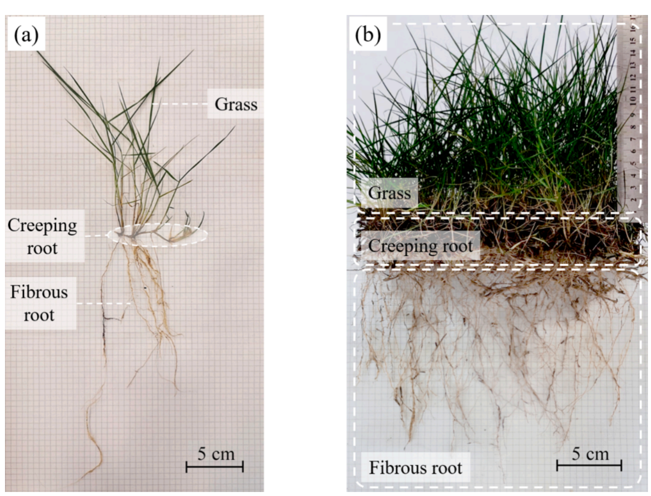

Fig. 3. Plant features of *Cynodon dactylon* (a) Single-root morphological trait (b) Group-root morphological trait.

root biomass, the initial root biomass retention ratio was set to 1. The coefficient k was obtained through regression analysis using a nonlinear exponential equation.

$$X_t/X_0 = e^{-kt} \tag{9}$$

where X_0 and X_t represent the initial dry mass of the root and the remaining dry mass at decay time t (g), respectively. The ratio X_t / X_0 denotes the root biomass retention ratio at time t (%). The parameter k is the root decay constant.

Using the root diameter and root length data of both creeping and fibrous roots, the roots were assumed to be homogeneous cylinders. The changes in root volume for creeping and fibrous roots over the decay period were then calculated. Subsequently, based on the soil depth where each root type was located, the root volume ratio R_v was determined at different decay times. Finally, $R_v(t)$ was obtained by analyzing the variation patterns of the root volume ratio over time.

3.4. Measurement and calculation of SWRC

In this study, an electronic tensiometer and a soil moisture sensor were used to measure the suction and volumetric water content of the soil at depths of 5, 20, and 30 cm, respectively. The volumetric water content measured by the soil moisture sensor was then converted into the degree of saturation, defined as the ratio of a given volumetric water content to the saturated water content. Finally, the soil–water retention curve (SWRC) was derived based on the relationship between the degree of saturation and matric suction.

3.5. Measurement and calculation of permeability coefficients

The wetting front is the interface between wetted soil and dry soil as water infiltrates. However, in this study, the initial soil water content made it more difficult to observe the wetting front interface compared to dry soil. Therefore, the position of the wetting front was determined by changes in data recorded by the soil moisture sensor. This indicates that when the soil moisture sensor detects a sudden and significant change in water content from the stable initial moisture state, the wetting front has advanced to the monitoring point (Chen et al., 2022; Lei et al., 2020).

At this point, the distance from the soil surface to the monitoring point is considered the wetting front advance distance Z_f. Meanwhile, once water begins to seep out from the bottom of the soil column, the Z_f represents the distance from the soil surface to the bottom of the column.

In this study, the Wetting Front Advancement Method (WFAM) proposed by Li et al. (2009) was used to derive the permeability coefficient curves, which describe the relationship between the permeability coefficient and matric suction at different soil depths and decay times:

$$k_u = \frac{[\theta(h_b, t_2) + \theta(h_b, t_1) - 2\theta_0]v^2\gamma_w(t_2 - t_1)}{2[s(h_b, t_1) - s(h_b, t_2) - v\gamma_w(t_2 - t_1)]} \tag{10}$$

Where k_u is the average permeability coefficient at t_1 and t_2 (cm/min); h_b is the distance from any moisture sensor to the soil surface (cm); $\theta(h_b, t_1)$ is the volumetric water content at h_b at t_1; $\theta(h_b, t_2)$ is the volumetric water content at h_b at t_2; θ_0 is the initial volumetric water content; γ_w is the heaviness of the water; v is the wetting front velocity (cm/min); $s(h_b, t_1)$ is the suction at h_b at t_1 (kPa); $s(h_b, t_2)$ is the suction at h_b at t_2 (kPa).

The saturated permeability coefficient, k_s, can be obtained from the permeability coefficient curve. Previous studies on soil permeability coefficient curves have shown that when the suction force is less than 1 kPa, the change in the permeability coefficient tends to stabilize, indicating that the soil is nearly fully saturated (Gallage et al., 2013). The permeability coefficient at a suction force of 0.1 kPa more accurately reflects the saturated permeability characteristics of the soil and provides high measurement accuracy is high (Rahimi et al., 2015). Therefore, the permeability coefficient corresponding to a suction force of 0.1 kPa in the permeability coefficient curve is considered the saturated permeability coefficient, k_s.

4. Test results and model validation

4.1. Root morphological trait of plants after decay

As the decay time increased, the biomass, diameter, length, and number of plant roots decreased compared to those of the undecayed roots (Table 2). The number of fibrous roots decreased from 434 to 154, representing a 65 % decline. The number of creeping roots decreased from 314 to 165, a reduction of 48 %. The diameter of fibrous roots decreased from 0.1–0.72 mm to 0.05–0.28 mm, reflecting a decline of 61 % to 68 %. The diameter of creeping roots decreased from 0.52–2.39 mm to 0.07–0.79 mm, indicating a decrease of 67 % to 75 %.

Statistical analysis revealed a significant difference in the range of root diameters between creeping and fibrous roots, with creeping roots being approximately twice as large as fibrous roots. During the decay process, the median diameter of both fibrous and creeping roots progressively decreased over time. By six months of decay, the median diameter was significantly lower than at one month, highlighting the substantial impact of decay duration on root diameter reduction. Compared to fibrous roots, creeping roots exhibited lower variation in diameter at 0 and 1 month, suggesting that creeping roots decay at a slower rate during the early stages. However, by the second month, root diameter variability in creeping roots increased sharply, indicating that significant decay begins between one and two months. After six months of decay, the advanced decomposition resulted in a narrower root diameter range, leading to reduced variability in both root types (Fig. 4).

The percentage of fibrous roots with a length between 0–5 cm increased with decay time, rising from 23.81 % at no decay to 79.09 % after six months of decay, an increase of 55.28 %. Conversely, the percentages of fibrous roots with lengths of 5–10 cm, 10–15 cm, and greater than 15 cm decreased by 18.61 %, 16.9 %, and 9.77 %, respectively.

Compared to healthy roots, root biomass decreased by 8.49 %, 18.53 %, 47.49 %, and 54.05 % at 1, 2, 4, and 6 months of decay, respectively (Table 2). By fitting the test results to Eq. (9), the root decay constant k is 0.13, with an R^2 value of 0.966. Additionally, the most significant decline in root biomass occurred between 2 and 4 months of decay, whereas the decrease in biomass was minimal between 4 and 6 months (Fig. 5).

4.2. Root volume ratio decay equation Rv(t)

The root volume ratio determines the proportion of roots in the soil, playing a crucial role in changes to soil pore structure due to the formation of root cavity channels during decay. The growth depth and characteristics of creeping and fibrous roots differ, and the root volume

Table 2
Summary of root length, root number, and root diameter data of *Cynodon Dactylon* at different decay times.

Decay time (month)	Root type					
	Fibrous root				Creeping root	
	Root length(cm)	Percentage of root number(%)	Root number	Root diameter(mm)	Root number	Root diameter(mm)
0	0–5	23.81	434	0.16–0.72	314	0.52–2.39
	5–10	44.62				
	10–15	20.82				
	>15	10.75				
	Root biomass(g) 25.9					
1	0–5	35.71	306	0.14–0.65	299	0.35–1.67
	5–10	36.26				
	10–15	19.48				
	>15	8.55				
	Root biomass(g) 23.7					
2	0–5	45.57	251	0.10–0.40	271	0.18–1.36
	5–10	34.50				
	10–15	13.55				
	>15	6.38				
	Root biomass(g) 21.1					
4	0–5	70.28	186	0.05–0.36	194	0.14–1.01
	5–10	23.04				
	10–15	5.07				
	>15	1.6				
	Root biomass(g) 13.6					
6	0–5	79.09	154	0.05–0.28	165	0.13–0.79
	5–10	16.01				
	10–15	3.92				
	>15	0.98				
	Root biomass(g) 11.9					

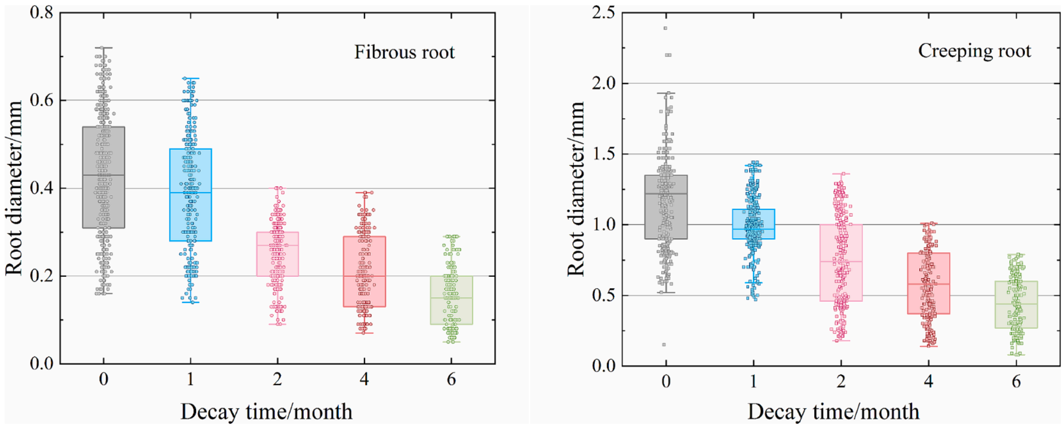

Fig. 4. Statistical results of root diameter (a) Fibrous root (b) Creeping root.

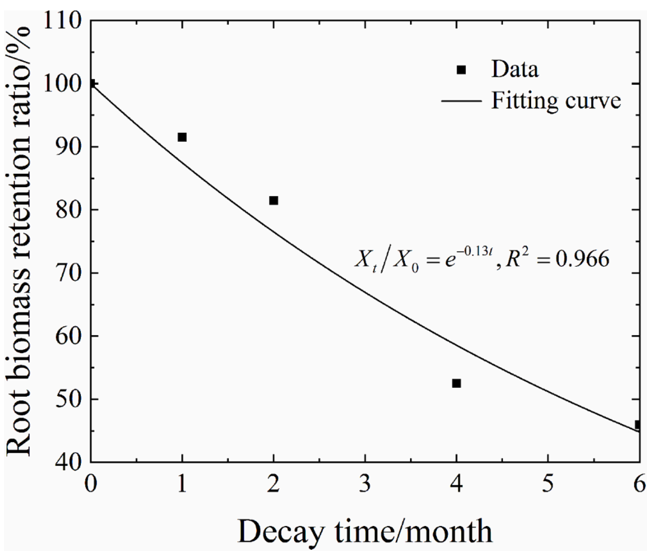

Fig. 5. Curve of root biomass retention ratio over decay time.

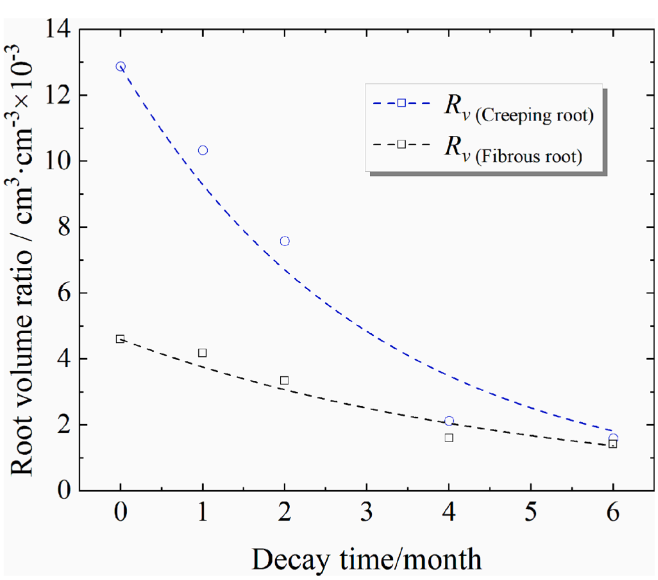

Fig. 6. Curve of root volume ratio over decay time.

ratio R_v over time was plotted for both root types (Fig. 6).

By fitting the root volume ratio and decay time data, the root volume ratio decay equation $R_v(t)$ was obtained:

$$R_v(t) = R_{v0}\exp(-Kt) \tag{11}$$

where R_v represents the root volume ratio (volume of plant roots per unit volume of the root-soil composite). When $t = 0$, the root volume ratio corresponding to the undecayed roots is expressed as R_{v0} (cm^3/cm^3), and K is the root volume ratio decay constant.

The top 5 cm of soil was dominated by creeping roots with a larger diameter, while the 20 cm depth provided a growth environment for fibrous roots. By comparing the decay patterns of the volume ratios of both root types, it was observed that creeping roots initially had a significant volume advantage, with their root volume ratio reaching 2.8 times that of fibrous roots. However, during the decay process, creeping and fibrous roots exhibited different characteristics: the volume ratio of creeping roots declined more rapidly over time, and its decay constant K was also larger than that of fibrous roots (Table 3). As the decay time increased to six months, the volume ratios of both root types converged (Fig. 6). The 30 cm depth consisted of bare soil with no root involvement, so the effect of root volume ratio decay was not considered at this depth.

4.3. Root decay causes the change of wetting front migration

The time it took for the wetting front to advance to the same soil depth decreased as the root decay time increased, and this change leveled off after four months of root decay. When the roots had decayed for six months, the time for the wetting front to advance at the same depth was more than halved compared to the soil with undecayed roots (Table 4).

By fitting the wetting front advance distance and time change data, it was found that they followed a power-function relationship (Fig. 7):

$$Z_f = a \times t^b \tag{12}$$

where Z_f is the wetting front advance distance (cm), t is the time (min), and a and b are the constants (Table 5).

Derivation of Eq. (12) and then take the logarithm as follows:

$$\log v = \lambda \times \log t + c \tag{13}$$

Table 3
Parameters of the root volume ratio attenuation equation.

Root type	Soil depth (cm)	R_{v0} (cm^3/cm^3)	K	R^2
Creeping root	5	0.0129	0.33	0.962
Fibrous root	20	0.0046	0.20	0.947

Table 4
Wetting front advance distance and corresponding moments at different times of root decay after precipitation.

Decay time (month)	Wetting front advance time to various depth sensors(s)			
	5c m	20c m	30c m	40c m
0	1379	5756	10,386	15,649
1	1201	4804	8760	14,023
2	993	3806	6764	11,864
4	709	2432	4683	9228
6	593	2277	4416	9120

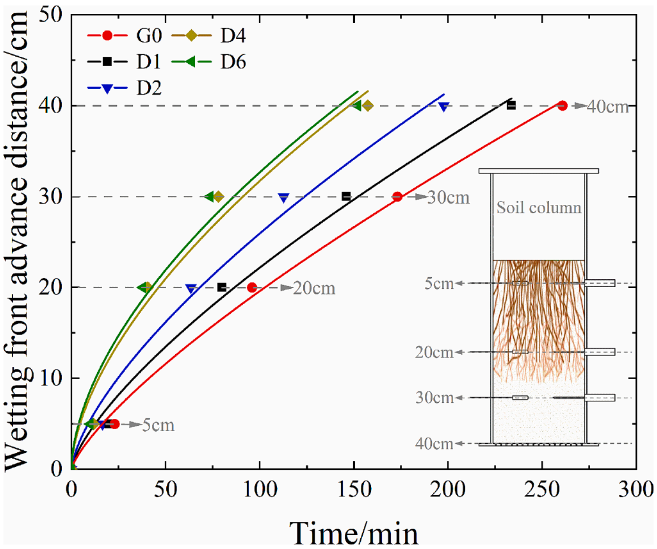

Fig. 7. Wetting front advance distance with time curve.

Table 5
Fitting parameters for wetting front advance distance as a function of time.

Decay time(month)	*a*	*b*	R^2
0	0.604	0.756	0.996
1	0.815	0.718	0.992
2	1.137	0.679	0.985
4	2.028	0.597	0.968
6	2.311	0.575	0.971

where v is the wetting front velocity (cm/min), t is the time (min), and λ and c are constants.

The curve of the wetting front velocity over time was obtained from Eq. (13) (Fig. 8). As the infiltration time increases, the wetting front velocity decreases. In other words, the wetting front velocity decreases with increasing depth (Huang et al., 2013; Lei et al., 2020). With increasing root decay time, the wetting front velocity became 1.2 to 2.3 times faster than in undecayed soil. The change in trend was more pronounced at shallower soil depths. When precipitation infiltrates deep into the lower layers of the soil column, the soil consists of bare soil without root involvement. Therefore, the soil properties were the same for each group, and eventually, the wetting front speed tended to be similar.

4.4. SWRC time-varying model validation

The time-varying SWRC model developed in this study has nine parameters, which were fitted using Eq. (4) based on the SWRC data at the time of root non-decay ($t = 0$) and calibrated for the parameters m_1, m_2, m_3, m_4 and p. Subsequently, we combined these with the parameters R_{v0} and K at different depths (Table 6) to predict the SWRC of the root-soil composite at depths of 5 cm and 20 cm at different decay times ($t =$ 1,2,4, 6). For the bare soil at a depth of 30 cm, the parameters R_{v0}, K, and p were set to 0 to fit the data of the G0 group and to verify the fit of this curve to the other four test datasets (D1, D2, D4, and D6).

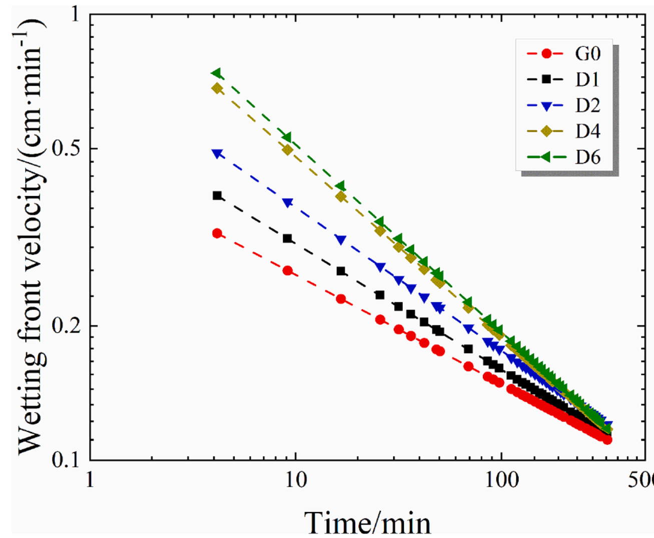

Fig. 8. Wetting front velocity with time curve.

At both 5 cm and 20 cm depths, under different decay times (D1, D2, D4, and D6, representing decay for 1, 2, 4, and 6 months, respectively), each predicted curve closely followed the overall trend of the measured data (Fig. 9a-b), indicating good agreement between the predicted and measured values (Table 7). The trend of each SWRC curve remained consistent: S_r decreased with increasing suction. When the suction force was less than the air-entry value (AEV), S_r changed minimally, resulting in a relatively flat curve. However, when the suction force exceeded the AEV, S_r entered an accelerated decline phase until reaching the residual S_r. As root decay progressed, the residual S_r of the soil at 5 cm and 20 cm gradually decreased, causing the curve to shift leftward while the downward section became progressively steeper. This pattern reflected a decrease in the AEV and a reduction in the soil's water-retention capacity. Notably, these trends were more pronounced at 5 cm (Fig. 9a-b). Thus, the shallower the soil, the greater the impact of root decay on its water-holding capacity. In contrast, for the bare soil at 30 cm, where the density and soil pore structure remained similar across groups, the variations in test data were minimal. As a result, the SWRC curve for group G0 closely aligned with those of the other groups (Fig. 2c).

4.5. Verification of time-varying permeability coefficient models

The void ratio e and saturated permeability coefficient (k_s at different soil depths, when the roots were not decayed ($t = 0$), were used to calibrate coefficient A using Eq. (7). The k_s value at 5 cm and 20 cm depths for the root-soil composite at different decay times ($t = 1,2,4, 6$) were predicted by incorporating the parameters root volume ratio R_{v0}) and attenuation coefficient K (Table 6). The k_s value for the bare soil at 30 cm was measured using the G0 group and compared with the test data from the other four groups (D1, D2, D4, and D6) to assess the degree of variation.

The predicted values calculated by the time-varying saturated permeability coefficient model at 5 cm and 20 cm depths showed a high degree of agreement with the measured data obtained from the permeability coefficient curves (Fig. 10a-b). In each group, k_s increased with decay time and decreased with soil depth. The soil k_s at 5 cm was 5.5 times higher than that of undecayed roots after six months of root decay and 2.1 times higher at 20 cm (Fig. 10a-b). The differences in test data between the groups were smaller for the bare soil at 30 cm

Table 6
Parameters of time-varying models.

Soil depth (cm)	Calibration parameters							Root volume ratio parameter		Bare soil's void ratio
	m_1	m_2	m_3	m_4	p	A (s/m)	k_s (m/s)	R_{v0}	K	e_0
5	0.578	1.973	7.162	0.980	1.7	4.60×10^{-3}	1.49×10^{-4}	0.0129	0.33	0.37
20	0.546	2.074	5.021	1.000	1	7.28×10^{-5}	7.28×10^{-5}	0.0046	0.20	0.37
30	0.519	2.080	3.514	1.114	—	3.30×10^{-4}	1.22×10^{-5}	—	—	0.37

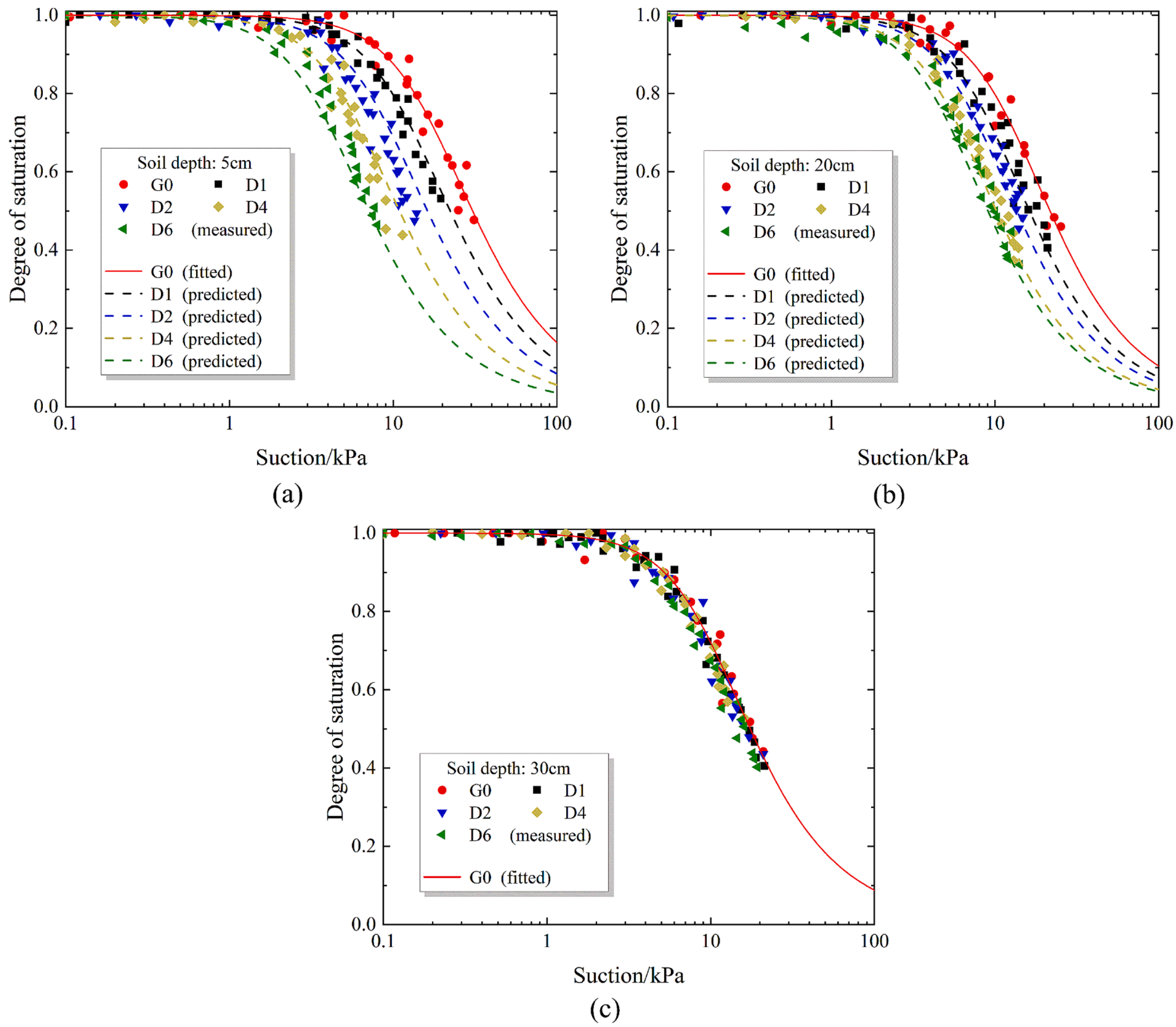

Fig. 9. Comparison between prediction and measurement of SWRC (a) 5 cm (b) 20 cm (c)30 cm.

Table 7
SWRC time-varying model predictions *R2* and *RMSE*.

Soil depth(cm)	Decay time (month)							
	1		2		4		6	
	R^2	*RMSE*	R^2	*RMSE*	R^2	*RMSE*	R^2	*RMSE*
5 cm	0.9197	0.0602	0.9356	0.0531	0.9142	0.0639	0.9432	0.0466
20 cm	0.9195	0.0633	0.9523	0.0484	0.9786	0.0343	0.9709	0.0345
30 cm	0.9851	0.0208	0.9683	0.0339	0.9747	0.0280	0.9728	0.0293

(Fig. 10a-b). The R^2 value of the model predictions was generally high ($R^2 > 0.96$), and the *RMSE* values were small (Fig. 10), indicating that the model was able to capture the changing rule of k_s at different depths under root decay and provided accurate prediction.

The SWRC time-varying model and the saturated permeability coefficient time-varying model were incorporated into Eq. (8) to predict

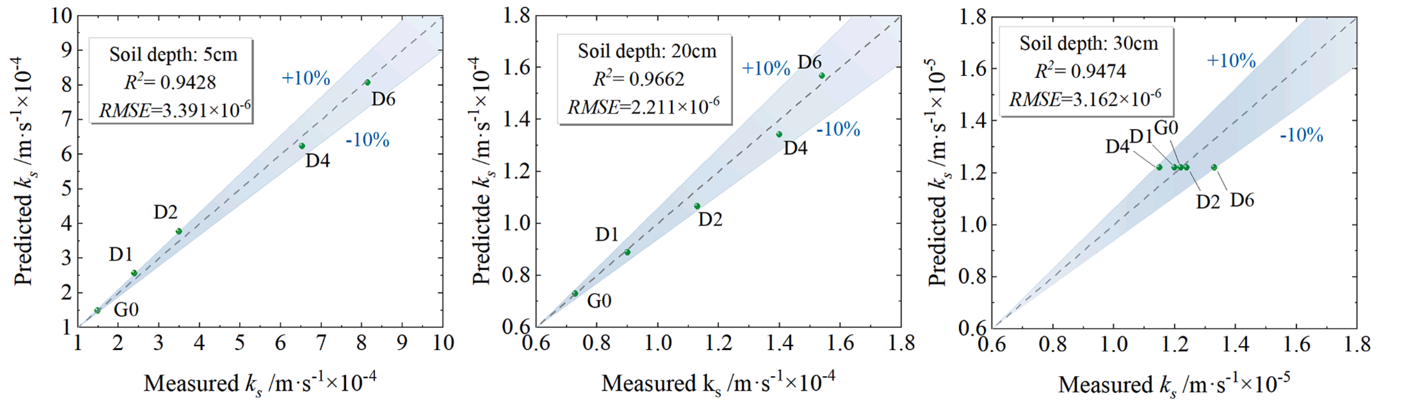

Fig. 10. Comparison between prediction and measurement of saturation permeability coefficient (a) 5 cm (b) 20 cm (c) 30 cm.

the permeability coefficient curves at different decay times (t = 1, 2, 4, 6). The overall trend of the measured data was well characterized by the predicted curves at both 5 cm and 20 cm depths (Fig. 11a–b). For the bare soil at 30 cm, the permeability coefficient curves of the G0 group were well-fitted to the test results of the other four groups.

The permeability coefficient spanned several orders of magnitude during the wetting process of the soil column and exhibited a staged decrease with increasing suction (Fig. 11). As suction increased from 0 kPa to the air entry value (AEV) (i.e., the suction level at which the permeability coefficient began to decline significantly), the soil water content remained close to saturation, causing the infiltration rate and permeability coefficient to remain nearly constant. From the AEV to the residual suction stage, the permeability coefficient decreased linearly with increasing suction. During this stage, the soil had a low water content, the water within the pores was significantly reduced, and the infiltration rate declined accordingly, resulting in a rapid decrease in the

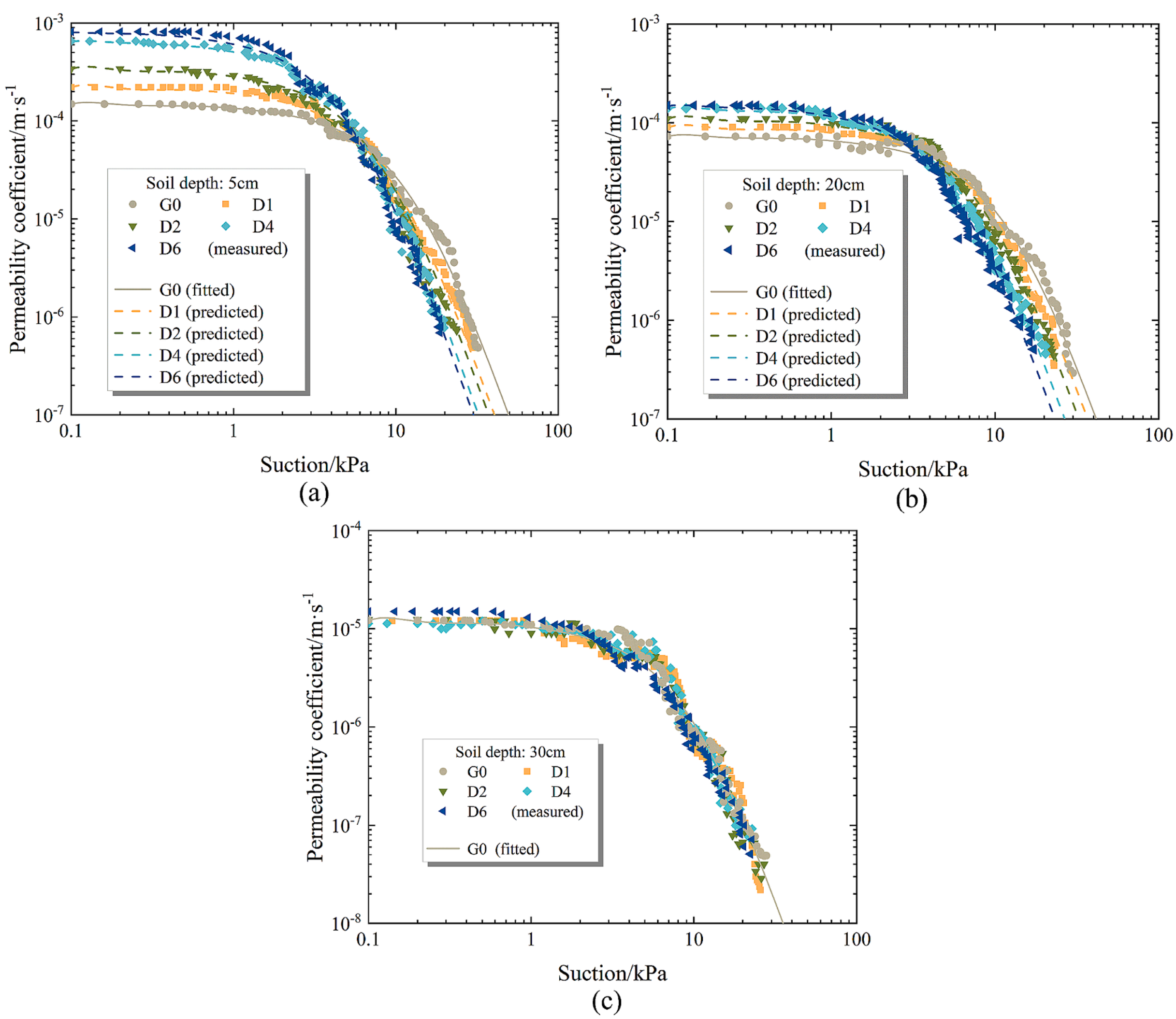

Fig. 11. Comparison between predicted and measured unsaturated permeability coefficients at different depths: (a) 5 cm, (b) 20 cm, (c) 30 cm.

permeability coefficient.

For root-bearing soils, differences in soil permeability properties were observed at different depths (5 cm and 20 cm) (Fig. 11a–b). In the first stage (i.e., the high suction range), the permeability coefficients at 5 cm soil depth were higher than those at 20 cm. However, in the second stage (i.e., the low suction range), the slope of the linearly decreasing section was steeper at 5 cm than at 20 cm. This indicates that the permeability coefficient at 5 cm decreased more rapidly with increasing suction, suggesting that the topsoil was more permeable than the deeper soil during root decay.

The R^2 value of the model predictions was generally high ($R^2 > 0.94$) (Table 8), while the *RMSE* values were consistently small, indicating that the models were well-fitted, exhibited high overall accuracy, and effectively captured the nonlinear changes in permeability coefficients with suction at different decay times.

5. Discussion

5.1. Root morphological traits of plants after decay

To quantify the degree of root decay, the root decay constant of 0.13 for *Cynodon dactylon* was obtained using Eq. (9) (Fig. 5). The root decay constant for grass plants that decay naturally through the bagging method ranged from 0.08 to 0.17 (Silver and Miya, 2001). These comparison constants indicate that this experiment is reliable for the artificial degradation and decay of roots using herbicides. Moreover, our findings indicate a negative exponential relationship between the root biomass retention ratio and decay time, implying that root decay is a gradual process (Fig. 5). In the initial stages of root decay, the discrepancy in weight loss rates is minimal due to the high root biomass density (Berg and McClaugherty, 2020), resulting in a small degree of decay. In the mid-stage of decay, the process is further accelerated by increased soil microbial activity (Wang et al., 2020), which is stimulated by the large amount of nutrients provided by the humus produced during root decay. However, the accumulation of residual lignin in the later stages leads to a slowdown in the rate of decay (Bryanin et al., 2018).

In general, the larger the root system's diameter, the lower the level of decay. However, in these tests, the average diameter of creeping roots was greater than that of fibrous roots. During the decay process, the rate of decrease in the average diameter of creeping roots was six percentage points higher than that of fibrous roots (Table 2). This indicates that the degree of decay in the *Cynodon dactylon* root system was inversely proportional to the root diameter. This may be because the xylem in fibrous roots is much less abundant than in creeping roots. During decay, the xylem of the roots decomposes earlier and more easily than the root bark. Additionally, creeping roots primarily grow within a depth of 5 cm from the surface, with some extending to the surface, whereas fibrous roots grow below 5 cm. Oxygen is more abundant near the surface, and the organic matter content (from decomposing residual turf) is relatively high, creating a favorable environment for aerobic microorganisms. These microorganisms rapidly break down the organic matter in the roots, accelerating the decay rate of creeping roots. As depth increases, oxygen levels decrease, and both microbial diversity and abundance decline significantly, leading to a slower decomposition rate in deeper fibrous roots.

Fibrous roots exhibited a characteristic trend in which the percentage of root numbers gradually declined as root length increased during decay (Table 2). This pattern suggests that, unlike the substantial change in the creeping root diameter during decay, fibrous roots decompose into multiple shorter fragments as they decay.

In this study, creeping and fibrous roots exhibited two distinct root architectures (Fig. 3a–b) and had different effects on soil structure during decay. As the roots decayed, the root volume ratio of both types decreased (Fig. 4). However, since creeping roots grew in the topsoil and had a diameter twice as large as the average diameter of fibrous roots (Fig. 4), they experienced a greater reduction in volume ratio (Table 3) and had a more pronounced impact on soil pore structure compared to fibrous roots.

This suggests that thicker, shallower roots are more effective at altering soil pore conditions and have a greater effect on soil hydraulic properties. In contrast, finer, vertically growing fibrous roots had a less significant effect on soil hydraulic properties, with their impact diminishing as depth increased.

5.2. Time-varying SWCC curve of soil under root decay

When SWRC curves are used to characterize the water-retention properties of soils, the trends of these curves at different depths and stages of decay vary (Fig. 9). At 5 cm and 20 cm depths, the AEV (the suction at the inflection point where the curve starts to decline) decreases as decay time increases, shifting the curve to the left. Meanwhile, the curve becomes steeper during the descending stage, indicating that suction is released more rapidly, which reduces the soil's water-retention capacity. This variation is associated with changes in soil pores caused by root decay.

Initially, the soil and roots are in close contact (Fig. 12a), but as the roots rot and become thinner, the root-soil pathways expand. The interconnection of channels formed by decayed roots increases soil porosity, resulting in a looser soil structure (Fig. 12b). During precipitation, water infiltrates the soil more quickly through these pores (Jiang et al., 2018), accelerating saturation and decreasing the soil's water-retention capacity.

Additionally, the more pronounced shift in the SWRC curve at 5 cm depth is due to the larger diameter and greater decay of creeping roots compared to fibrous roots, which increases topsoil porosity. For instance, after six months of root decay, the porosity of the topsoil becomes significantly larger due to the decomposition of creeping roots (Fig. 12b). Therefore, creeping roots growing in the topsoil are more likely to affect the soil's water-retention capacity during decay.

5.3. Time-varying law of soil permeability under root decay

Water infiltration properties can be analyzed indirectly using the wetting front velocity as an observation index, while the permeability coefficient curve and the saturated permeability coefficient directly reflect soil permeability capacity. In each group, the wetting front velocity decreased continuously. This decrease results from increasing soil water content and the subsequent reduction in suction, leading to a lower hydraulic gradient (Rubio et al., 2022). However, differences in wetting front velocity were observed between the groups after root

Table 8
Predictions of the permeability coefficient prediction model R^2 and *RMSE*.

Decay time(month)	5 cm		20 cm		30 cm	
	R^2	*RMSE*	R^2	*RMSE*	R^2	*RMSE*
0	0.9466	1.11×10^{-6}	0.9432	7.22×10^{-7}	0.9432	7.22×10^{-7}
1	0.9772	5.49×10^{-7}	0.9579	7.83×10^{-7}	0.9579	7.83×10^{-7}
2	0.9551	3.87×10^{-6}	0.9636	6.71×10^{-7}	0.9636	6.71×10^{-7}
4	0.9643	1.50×10^{-6}	0.9476	9.45×10^{-7}	0.9476	9.45×10^{-7}
6	0.9772	5.02×10^{-7}	0.9897	4.56×10^{-7}	0.9897	4.56×10^{-7}

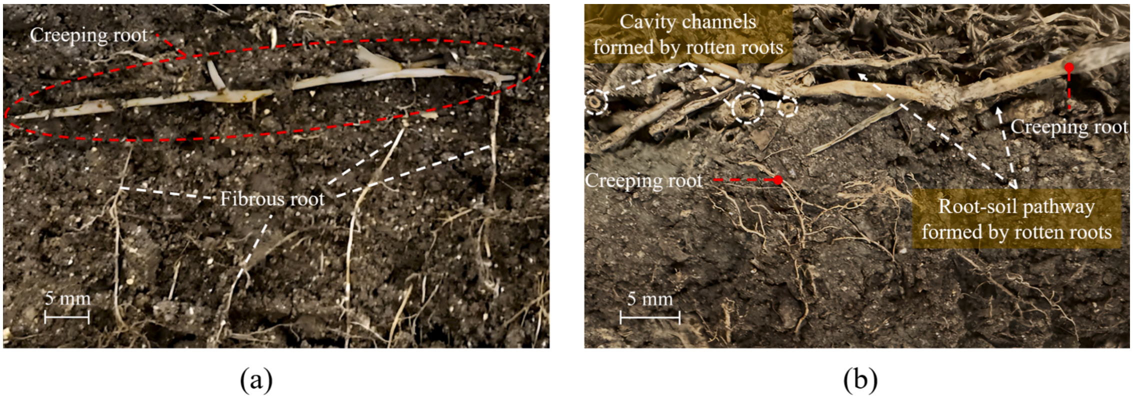

Fig. 12. Changes of soil structure before and after root decay (a) Growth of fresh roots in soil (b) Soil macropores are formed by root decay.

decay (Fig. 8). The shallower the soil, the greater this difference. This depth-dependent variation aligns with the characteristic changes in the saturated permeability coefficient (Fig. 10).

Therefore, the changes in the permeability properties of the topsoil during root decay are more significant than those in deeper soil layers. This also results in a more rapid change in the permeability coefficient of the topsoil (Fig. 11). For the same range of suction variations, topsoil water content changes more rapidly. Since the infiltration rate is closely related to water content, a faster change in water content leads to a faster change in infiltration rate, ultimately resulting in a more rapid change in the permeability coefficient with suction.

The permeability properties changed over time with root decay. This is because, as roots decay, they not only darken and thin but also experience internal degradation, with the xylem at the center of the root rotting first. This process leaves only the outer root bark, forming cavity channels in the soil. These cavity channels began to increase steadily after one month of root decay. By the fourth month, the roots had decomposed extensively into humus (Fig. 13a–b), which connected the cavity channels to macropores, allowing water to preferentially flow through these macropores and thus accelerating water infiltration.

Additionally, the high permeability exhibited by the topsoil is related to the decreasing root biomass with depth. As decay time increased, shallower soil layers experienced greater root decay, leading to the formation of more macropores and a faster infiltration rate. Creeping roots growing in the topsoil have thicker xylem than fibrous roots, and the cavity channels created by their decay are wider, further enhancing the permeability of the topsoil.

The pattern of root biomass change over time follows the same trend as the permeability change of the root-soil composite. The change in permeability properties is determined by the decay pattern of the root volume ratio. Permeability changed uniformly during the first two months of decay, with the most significant changes occurring between the second and fourth months, after which it stabilized. This is because the permeability of the root-soil composite is closely related to root biomass (Jotisankasa and Sirirattanachat, 2017). The more severe the root decay, the greater the soil permeability.

6. Conclusions

In this study, considering the time-varying effect of root decay on the hydraulic properties of the root-soil composite, time-varying models of hydraulic parameters that account for changes in soil porosity were established. The time-dependent variations in hydraulic properties were obtained through one-dimensional soil column infiltration tests, and the reliability of the models was verified. This study reveals the evolution mechanism of root decay on the hydraulic properties of the root-soil composite and leads to the following conclusions:

(1) The study captures the evolution of soil pores during root decay and establishes hydraulic parameter models. These include a time-varying soil water retention curve (SWRC) model, which reflects changes in the soil's water-retention capacity; a time-varying saturated permeability coefficient model, $K_s(t)$; and a permeability coefficient prediction model, which accounts for changes in the soil's permeability capacity. These models effectively represent the impact of soil pore evolution on the time-dependent hydraulic properties of the root-soil composite following root decay.

(2) Roots begin to decay after the plant dies. As decay progresses, the biomass, number, length, and diameter of the roots decrease until they eventually stabilize. The relationship between the root biomass retention ratio and decay time follows a negative exponential trend, with a root decay constant of 0.13. The root volume ratio exhibits a negative power function relationship with decay

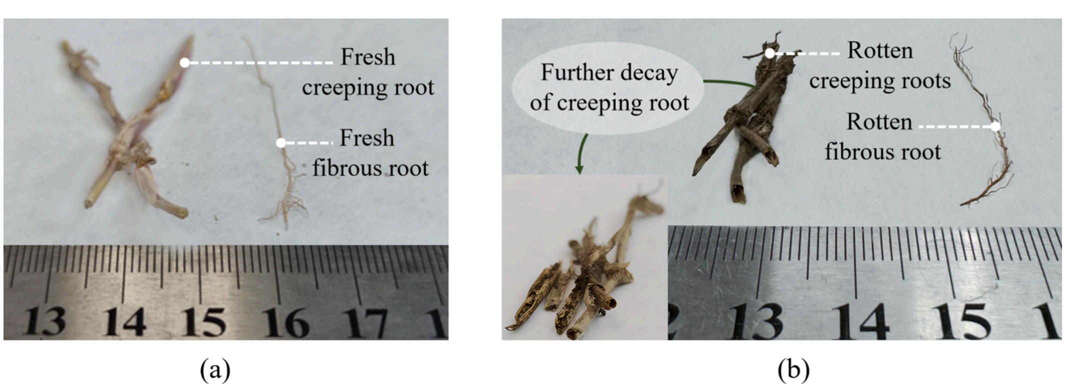

Fig. 13. Comparison of creeping and fibrous roots before and after root decay (a) Fresh roots (b) Decayed roots.

time, with a decay constant of 0.33 for creeping roots at a depth of 5 cm and 0.20 for fibrous roots at a depth of 20 cm. These findings suggest that creeping roots decay more rapidly and have a greater impact on the pore structure of the topsoil.

(3) Cavity channels and macropores formed by decayed roots increase soil porosity, alter soil structure, and influence permeability and water-retention capacity. The average diameter of creeping roots is approximately twice that of fibrous roots, leading to larger cavity channels in decayed creeping roots. This results in significant structural changes in the topsoil, making the time-dependent variations in hydraulic properties more pronounced.

(4) As root decay progresses, soil porosity increases, leading to a decrease in the air-entry value (AEV) of the soil water retention curve (SWRC), shifting the curve to the left. The curve becomes steeper in the decline phase, causing suction force to be released more rapidly, reducing residual saturation and diminishing the soil's water-retention capacity. Notably, due to the distinct root architectures of creeping and fibrous roots, this effect is more pronounced in the topsoil.

(5) Within six months of root decay, the wetting front velocity during precipitation was 1.2–2.3 times higher than in undecayed soil. The saturated permeability coefficient during root decay decreased with increasing soil depth. The saturated and unsaturated permeability coefficients at a depth of 5 cm were particularly pronounced; at six months of decay, the saturated permeability coefficient at 5 cm was 5.5 times higher than that of undecayed soil, whereas at 20 cm, it was 2.1 times higher. Consequently, topsoil exhibited greater permeability during root decay. The variation trends in permeability properties were consistent with those observed in root biomass decay.

CRediT authorship contribution statement

Suyun Meng: Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Tongyun Zhang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guoqing Zhao:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Shiwei Hou:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was funded by the National Natural Science Foundation of China [No. 52208356]. The authors declare no conflicts of interest. We are also grateful to the editor and the two anonymous reviewers who offered insightful comments leading to the improvement of this paper.

Data availability

Data will be made available on request.

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